Iberdrola Datathon 2026 — Project Overview

1. Goal

Identify **where to build new ultra-fast EV charging stations** and **which existing gas stations to upgrade** along Spain's interurban road backbone, so the network is ready for the 2027 EV demand scenario. The output is a deployment map with exact coordinates, charger counts per site, and the electrical grid reinforcements needed to support them.

The project is motivated by the **EU Alternative Fuels Infrastructure Regulation (AFIR, 2023/1804)**, which mandates fast-charging pools every "**60 km** along the TEN-T core network with minimum **400 kW**" (Europa.eu) starting from 2025, and full coverage of the comprehensive network by 2027.

2. Datasets

All raw data lives in *data/raw/* , is standardized to **EPSG:25830** (a standardized metric coordinate reference system used in Spain) in *data/standardized/* , and is fused into a single point-based backbone in *data/processed/backbone_foundation.parquet*.

Dataset	Source	Why it's used	How it's used
Roads (KMZ)	MITMA / Ministry of Transport	Defines the interurban "backbone" network where AFIR applies.	Discretized into points every 200 m (**discretize_backbone roads**) to create ~150k candidate demand nodes.
Traffic intensity	DGT Mapa de Tráfico (road segment AADT)	Proxies demand: more vehicles → more charging sessions needed.	Joined to backbone points via a 50 m buffer with neighborvalidation to avoid overpass "cross-talk"; yields **total_max** (total daily max) and **medio_max** (medium-distance share).
Existing EV chargers (XML)	MITECO / public registry	Supply baseline — we must not double-count sites that already exist, and the largest observed site sets the sitesize cap.	Loaded as fixed always-open sites with current **charger_count** as a lower bound; the empirical maximum is reused as **MAX_CHARGERS_PER_SITE** .
Gas stations (JSON)	CNMC / Ministry of Industry fuel registry	Candidate upgrade sites — they already have land, access, and grid connections.	Loaded as binary candidates (0/1) the optimizer may choose to convert to charging sites.
Electrical grid capacity (XLSX)	Endesa, Iberdrola, Viesgo public capacity maps	Feasibility filter: a site is useless without grid power, and capacity drives the gridstatus classification.	Each candidate is mapped to its nearest substation (≤ 10 km); its charger count is capped by **capacity_kw** , and the capacity/demand ratio tags each site as Sufficient / Moderate / Congested.
Vehicle registrations	DGT (2015–2025 monthly)	Drive the EV share used in demand scaling.	Monthly EV and non-EV series are modeled with **SARIMAX** with order=(0, 0, 2) & seasonal_order=(0, 0, 2, 12)

3. Assumptions & Hyperparameters

All values below are defined in the notebook (*notebooks/02_backbone_analysis.ipynb*), *config.toml*, and at the top of *scripts/processing/optimize_grid_aware_placement.py*. They combine real-world assumptions (regulation, industry standards) with constants that are now increasingly **derived from the data itself** rather than hard-coded.

Parameter	Value	Meaning & Rationale
EV traffic share (ev_traffic_pct)	Forecast (0.134493 for 2027)	Share of 2027 vehicle registrations expected to be EVs, computed from the SARIMAX forecast (vehicle_registrations_2027['pct_ev_registrations'])
Share needing a charge (need_charge_pct)	data-driven	Ratio medio_max / total_max across backbone traffic. Approximates the fraction of traffic on medium-distance trips that would need a fast-charge stop.
Sessions per charger per day (sessions_per_charger)	24	Throughput of one 150 kW charger assuming ~45 min sessions over an 18 h effective service window.
Min chargers per site (MIN_CHARGERS_PER_SITE)	4	AFIR minimum per charging pool.
Max chargers per site (max_chargers_per_site)	data-driven (max(charger_count) of existing sites)	Upper bound anchored to the largest real Spanish site, so the model can't prescribe something larger than what operators have actually deployed.
Charger power (power_per_charger_kw)	150 kW	AFIR minimum plug power for the TEN-T comprehensive network.
Coverage radius (coverage_threshold_m)	30 000 m	AFIR mandates ≤ 60 km between pools; a 30 km service radius guarantees that gap end-to-end while keeping candidate neighborhoods tractable.
Substation reach (substation_threshold_m)	10 000 m	Beyond 10 km, candidate is flagged as "No Grid Access".
Min supply/demand ratio (MIN_SUPPLY_DEMAND_RATIO)	1.0	Every demand point must be fully served by its local supply.
Coverage penalty (penalty_coverage)	1×10^6	Cost of leaving a demand point uncovered to adhere to AFIR compliance.
Supply penalty (penalty_supply)	1×10^4	Cost per missed charging session.
Grid-upgrade penalty (penalty_grid_upgrade)	1×10^2	Cost per kW of substation reinforcement. Kept low so the model prefers upgrading the grid over leaving corridor gaps.
Road sampling interval (sampling_interval_m)	200 m	Balance between spatial precision and tractability (~150k points nationwide).
Traffic buffer (buffer_radius_m)	50 m	Captures the actual carriageway while excluding parallel roads.
Backbone sampling step (SAMPLING_STEP)	1 or 25	1 = evaluate every backbone point (full precision); 25 = every 5 km (speed mode for grid-aware variant).
Forecast horizon	24 months	SARIMAX projects EV and non-EV registrations 2 years forward, capturing the full 2027 window used for ev_traffic_pct.

r

4. Optimization Model

The placement problem is framed as a **Capacitated Maximal Covering Location Problem (CMCLP)**, a operations-research formulation where we decide which sites to open and how to size them so that as much demand as possible is served within a fixed travel radius. It is solved as a **Mixed-Integer Linear Program** using **PuLP** with the **CBC solver** (in the notebook) and **SciPy's milp** with the **HiGHS backend** (in the full-scale script, which uses a linear relaxation for speed over 150k demand points).

The optimizer chooses, for each candidate site, whether to open it and how many 150 kW chargers to install there. Three types of candidates feed the model: the **existing charging stations** (fixed as always-open, so coverage is counted correctly), the **gas stations** (primary upgrade targets since they already have land, access, and a grid connection), and **greenfield points** (synthetic candidates placed on backbone segments that are either too far from any existing site or where the nearby supply cannot meet demand — this prevents the model from becoming infeasible in rural corridors). These candidates are produced by **generate_smart_candidates**(...) using the backbone, chargers, gas stations, plus the data-driven EV share, charge-need ratio, coverage threshold, site-size cap, and sessions-per-charger.

The objective is to **minimize the number of new stations and chargers while fully complying with AFIR coverage**. This is encoded through a hierarchy of penalties: leaving a stretch of road uncovered is heavily penalized (reflecting the 60 km AFIR rule), undersupplying demand is penalized moderately, and reinforcing the electrical grid is penalized only lightly. The ordering is deliberate. The model would rather upgrade a substation than leave a corridor gap, and would rather add chargers than open a whole new site.

Several real-world constraints shape the solution. Every point on the road backbone must have at least one open charging site within **30 km** (the updated coverage radius). Every site that opens must host between 4 and maximal 30 chargers per site. The total power drawn by chargers at each substation cannot exceed that substation's available capacity, unless the model explicitly flags a grid upgrade. And any candidate farther than **10 km** from the nearest substation is automatically excluded, since connecting it would be economically unreasonable.

The optimizer called **solve_grid_aware_optimization**`(...)` returns a candidate-level table and a per-substation gridslack vector. The table, written *to data/processed/grid_aware_optimized_sites.parquet*, tells us for each site whether to open it (is_open), how many chargers it should host (final_n), how many need to be added versus what exists today (added_chargers), and which substation it draws from. The grid slacks indicate how many kilowatts of reinforcement each substation needs.

5. Output Construction

The raw optimizer output is post-processed in the notebook into three business-ready CSVs written to *data/outputs/,* plus an interactive map.

Per-station enrichment. Each recommended site receives an **estimated_demand_kw** (**final_n** × 150), its latitude / **longitude**, a human-readable **location**_id (e.g. IBE_001 , IBE_002 …), and the nearest **route_segment** name via a spatial join against the backbone. The substation it draws from is merged back in with its company (distributor) and **capacity_kw** .

Grid status classification. Per substation, we aggregate **estimated_demand_kw** across all sites assigned to it and compute ratio = **capacity_kw** / **estimated_demand_kw**. The ratio is bucketed into three traffic-light categories (Sufficient, Moderate, and Congested) which flow through to both the deployment file and the friction-point file. Sites whose substation is not Sufficient are flagged as bottlenecks needing grid-reinforcement coordination with the relevant distributor.

File 1 — KPI scorecard (File 1.csv). A single-row summary with total_proposed_stations , total_existing_stations_baseline , total_friction_points , and total_ev_projected_2027 . Meant as the executive headline.

File 2 — Station deployment map (File 2.csv). One row per proposed new station: location_id , latitude , longitude , route_segment , n_chargers_proposed , grid_status . This is the operational "where to build" output.

File 3 — Grid friction points (File 3.csv). One row per bottleneck site (Moderate or Congested): bottleneck_id (e.g. FRIC_001), latitude , longitude , route_segment , distributor_network , estimated_demand_kw , grid_status .

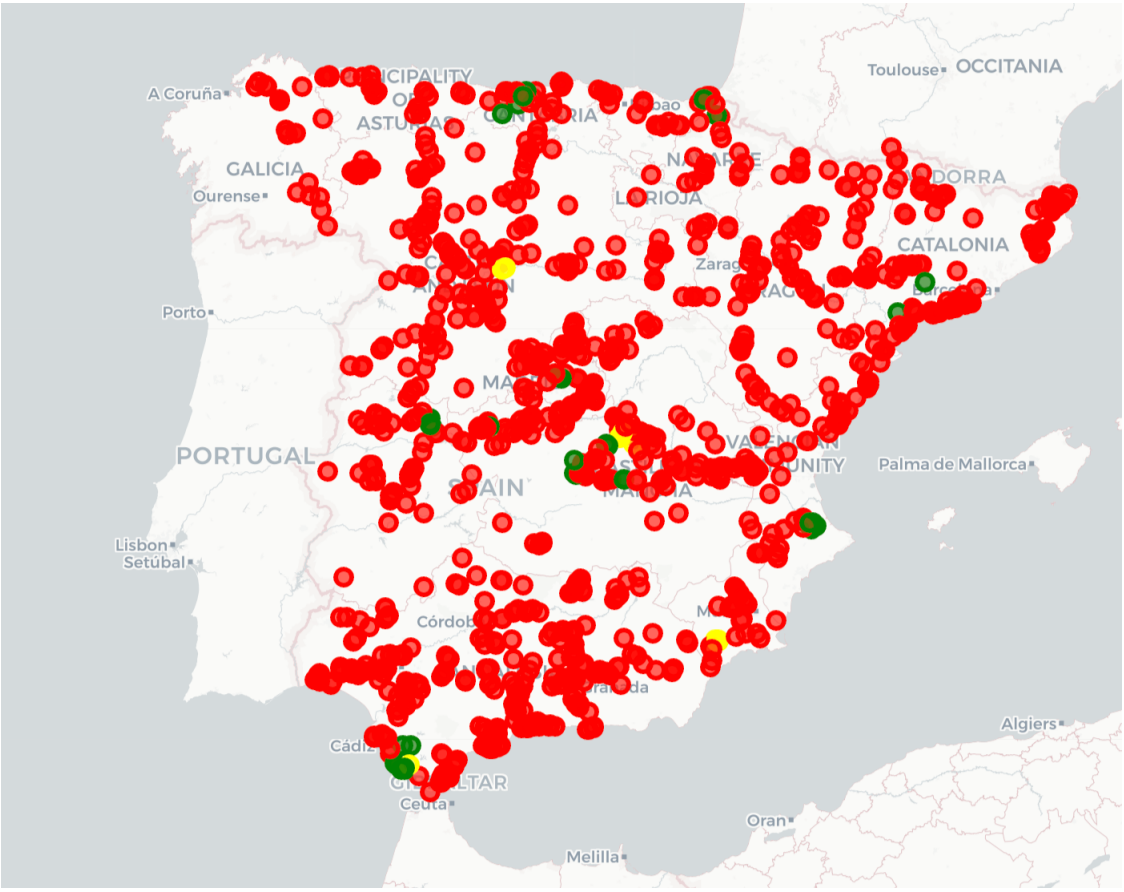
Final visualization. A Folium map (grid_status_map.html) plots every proposed site colored by grid_status — green for Sufficient, yellow for Moderate, red for Congested — with hover tooltips showing the route segment, charger count, status, and coordinates.

6. Results & Discussion

Running the full pipeline end-to-end for the 2027 horizon produces the following headline KPIs:

Metric	Value
Total proposed new stations	**1 215**
Existing stations (baseline)	**1 298**
Grid friction points	**1 215**
Total EVs projected by 2027	**655 995**

The map below visualizes every proposed station, colored by the grid-status classification introduced in Section 5 — green = *Sufficient*, yellow = *Moderate*, red = *Congested*.



The recommendation adds **1 215 new stations**, nearly doubling today's baseline of 1 298 which is consistent with the proportional jump in EV stock projected for 2027 (≈ 656 000 vehicles). The striking finding, however, is that the number of friction points exactly equals the number of proposed stations: **100 % of the new sites fall on substations that are either Moderate or Congested relative to the new load**. Visually, the map is dominated by red along every major interurban corridor, with *Sufficient* (green) sites confined to a handful of dense urban hubs (Madrid, Barcelona, Bilbao, Murcia/Alicante, and the Cádiz–Gibraltar corridor) where multiple high-voltage substations already serve redundant demand. This means the 2027 rollout is **grid-bound, not site-bound**: the physical locations the optimizer wants to use are available and economically reasonable, but the MV/HV network feeding them is the binding constraint. Operationally, this reframes the deliverable File 3.csv (friction points) is not a secondary artifact but the primary coordination document aiming to reinforce the grid concurrently with station construction. Without that parallel workstream, the AFIR 60 km compliance target is met on paper but not in practice, because congested substations will cap the usable charging throughput well below the proposed final_n.

7. Requirements

- Python ≥ 3.13, managed with UV
- Core Libraries: geopandas, shapely, scipy, pulp, joblib, pmdarima, statsmodels, polars, folium, branca
- CBC solver (budled with PuLP) or HiGHS (bundled with SciPy ≥ 1.11
- Access to GCP bucket for the pre-built silver layer *python3 scripts/sync_clouds.py*

8. Pipeline Summary

```
raw/ ──▶ 01_acquisition.py ──▶ standardized/ ──▶ 03_processing.py ──▶ backbone_foundation.parquet
│                                                                    ├──▶ SARIMAX forecast (EV
+ non-EV registrations → 2027 EV share)
                                                                    │
                                    └──▶ optimize_grid_aware_placement.py
                                                                          │
                                          ├──▶ grid_aware_optimized_sites.parquet
                                                                                  │
                                                          └──▶ output construction
                                                                  ├──▶File1.csv      (KPI
                                                                  scorecard)
                                                                  ├──▶File2.csv
                                                                  (deployment map)
                                                                  ├──▶  File3.csv   (grid
                                                                  friction points)
                                                                  └──▶
                                                                  grid_status_map.html
```

The final deliverable is a deployment map answering three questions: **where** to put each station, **how many** chargers, and **which substations** need reinforcement to power them.